Romanian palatalization without tears

In Romanian, coronal and dorsal obstruents become palatalized when followed by certain suffixes beginning with a front vowel, such as noun plurals in *-i* (NB: /i/ ⇝ [ʲ] word-finally).

(1)	rus	ruʃʲ	‘Russian(s)’	brad	brazʲ	‘fir(s)’	munte	muntsʲ	‘mountain(s)’
	prost	proʃtʲ	‘fool(s)’	duŋgə	dundʒʲ	‘stripe(s)’	pisikə	pisitʃʲ	‘cat(s)’

Some details of these *palatalization* mutations—the precise phonetic form of the palatalized consonants, nasal place assimilation in ‘stripes’, etc.—are straightforwardly handled by a series of ordinary phonological rules or constraints. However, additional observations have led some (e.g., Chitoran 2001) to posit palatalization is morphophonological in part. Palatalization exhibits *non-derived environment blocking*: it does not apply when the obstruent target and front vowel trigger are tautomorphemic and underlyingly adjacent; e.g., [kilogram] ‘id.’, [gimpe] ‘thorn’. But there are also some derived exceptions, such as some agent nominals in *-ist*; e.g., [fok-ist] ‘stoker’, [taŋk-ist] ‘tank driver’, [drog-ist] ‘druggist’, [jog-ist] ‘yogi’; cf. [kajak, kajatʃ-ist] ‘kayak(er)’, [stɨŋg, stɨŋdʒ-ist] ‘left(ist)’. Steriade (2008) further claims nouns selecting plural allomorphs triggering palatalization (e.g., *-i*) tend to select palatalizing allomorphs of derivational suffixes—like the homophonous *-i* verbalizer as in (2a)—and nouns which select non-palatalizing plural allomorphs (e.g., *-uri*) tend to select non-palatalizing verbalizer allomorphs (e.g., *-a* or *-ui*) as in (2b).

(2)	a.	kolak	kolatʃʲ	‘kalach(es)’	ɨŋkolatʃʲ	‘to coil’
		verde	verzʲ	‘green(s)’	ɨnverzʲ	‘to become green’
	b.	fok	fokurʲ	‘fire(s)’	ɨnfoka	‘to fire up’
		tɨrg	tɨrgurʲ	‘market(s)’	tɨrgui	‘to go shopping’

Taking the above observations into account, we consider the rules governing palatalization and their productivity. Productivity analyses are conducted using a lexicon extracted from Wiktionary and the DEX dictionary (Academia Română 1998) filtered by frequency (Goldhahn et al. 2012) to simulate realistic lexical knowledge (e.g., Kodner 2019). Below we give the type frequencies of noun plural palatalization (*n*: # of stems eligible for palatalization, *e*: # of stems which are exceptions to palatalization) for the 1,000 most-common nouns in our sample, with majority-rules (Nida 1949:14) and Tolerance Principle (Yang 2005, 2016) calculations, showing all coronal- and dorsal-obstruent-final stems when followed by heterosyllabic *i* or *e*. Under both thresholds, there is productive palatalization of *c* and *g* before *i*, *e* and of *t*, *d*, and *s* before *i*.

(3)

	n	*e*	majority?	tolerated?		*n*	*e*	majority?	tolerated?
…c-i…	28	0	✓	✓	*…c-e…*	14	0	✓	✓
…g-i…	17	0	✓	✓	*…g-e…*	3	0	✓	✓
…t-i…	207	0	✓	✓	*…t-e…*	229	229	✗	✗
…d-i…	23	0	✓	✓	*…d-e…*	36	36	✗	✗
…s-i…	11	0	✓	✓	*…s-e…*	44	44	✗	✗
…z-i…	2	1	(n.a.)	(n.a.)	*…z-e…*	48	48	✗	✗
…st-i…	4	0	✓	✓	*…st-e…*	0	0	(n.a.)	(n.a.)
…ct-i…	3	0	✓	✓	*…ct-e…*	22	22	✗	✗

These palatalization and allomorphy facts derive directly from the narrow phonology. Following *inalterability as prespecification* (e.g., Inkelas and Cho 1993, Kiparsky 1993, Buckley 1994, Inkelas 1995, Inkelas et al. 1997, Gorman 2026), underspecified segments are transparent to palatalization while prespecified segments are exempt (Rasin 2025); the rules of exponence are sensitive to the same specification. For sake of brevity, we focus on masculine nouns ending in *k* or *g*, though the basic account generalizes also to coronals. To implement the desired feature-filling behavior, we use the notion of unification (Bale et al. 2014). Rules (4–5) maps underspecified /K, G/ to [tʃ, dʒ] (resp.) before front vowels and to [k, g] elsewhere. While these rules intensionally target many segments, unification does not overwrite existing specifications, so fully-specified segments like /k, g/ are inalterable w.r.t. these rules.

(4) Dorsal palatalization:

$$\begin{bmatrix} -\textsc{Syllabic} \\ +\textsc{Consonantal} \\ -\textsc{Sonorant} \\ -\textsc{Approximant} \\ -\textsc{Nasal} \\ \ldots \end{bmatrix} \sqcup \left\{ \begin{matrix} +\textsc{Coronal} \\ -\textsc{Dorsal} \\ -\textsc{Anterior} \\ +\textsc{Distributed} \\ +\textsc{Strident} \end{matrix} \right\} / \_\_ \begin{bmatrix} +\textsc{Syllabic} \\ +\textsc{Front} \end{bmatrix}$$

(5) Dorsal default fill-in:

$$\begin{bmatrix} -\textsc{Syllabic} \\ +\textsc{Consonantal} \\ -\textsc{Sonorant} \\ -\textsc{Approximant} \\ -\textsc{Nasal} \\ \ldots \end{bmatrix} \sqcup \left\{ \begin{matrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{Continuant} \end{matrix} \right\}$$

We now turn to allomorph selection. Assuming inside-out spell-out (e.g., Bobaljik 2000), we propose that spell-out of the noun plural suffix Num[Pl] and verbalizer *v* is conditioned on the underlying phonological form of the root. Roots prespecified for −Coronal, +Dorsal, and −continuant select the non-palatalizing allomorphs *-uri* and *-a*. On the other hand, palatalizing roots lack these specifications and thus fall through to the elsewhere allomorph *-i*, with a rather similar pattern governing spell-out of the verbalizer (*v*) allomorphs *-a* and *-i*.

(6) $\text{Num[Pl]} \iff \begin{cases} \text{/-uri/} & / \begin{bmatrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{continuant} \end{bmatrix} \_\_ \\ \text{/-i/} & \text{elsewhere} \end{cases}$ (7) $v \iff \begin{cases} \text{/-a/} & / \begin{bmatrix} -\textsc{Coronal} \\ +\textsc{Dorsal} \\ -\textsc{continuant} \end{bmatrix} \_\_ \\ \text{/-i/} & \text{elsewhere} \end{cases}$

The interaction between allomorph selection and palatalization is illustrated in the derivations below, which begin just after root spell-out.

(8)

UR:	fok	fok⁀Num[Pl]	koloK	koloK⁀Num[Pl]	miloG	miloG⁀*v*	duŋG	duŋG⁀Num[Pl]
Rule (6):		fokuri		koloKi				duŋGi
Rule (7):						miloGi		
Rule (4):				kolatʃi		milodʒi		dundʒi
Rule (5):			kolak				duŋg	
Desyllabification:		fokur^j		kolatʃ^j		milodʒ^j		dundʒ^j
SR:	fok	fokur^j	kolak	kolatʃ^j	milog	milodʒ^j	duŋg	dundʒ^j

We tentatively propose that plural choice and derivational behavior could appear to align simply because both are conditioned by the same phonological context. This possibility casts doubt on Steriade's *inflection dependency* account; in other words, it shows the supposed effect can be generated without recourse to the "paradigm referral" analysis Steriade proposes. Further doubts about the reality of this effect arise in (9), where we test for a correlation between a noun's plural-class and palatalization of its derived verb's root consonant; the Fisher exact test finds no such correlation for dorsals, coronals, or the two classes combined.

(9)

	c, g				*t, d, s*				*c, g + t, d, s*			
	tʃ, dʒ	*k, g*	OR	*p*	*ts, z, ʃ*	*t, d, s*	OR	*p*	*tʃ, dʒ, ts, z, ʃ*	*k, g, t, d, s*	OR	*p*
i, e	10	24	1.53	.728	6	58	1.29	1.00	16	82	1.40	.613
uri	3	11			2	25			5	36		

References

Academia Română, ed. 1998. *Dicționarul explicativ al limbii române*. Univers Enciclopedic, 2nd edition.

Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: A formal analysis of feature-changing and feature-filling rules. *Lingua* 148:240–253.

Bobaljik, Jonathan David. 2000. The ins and outs of contextual allomorphy. *University of Maryland Working Papers in Linguistics* 10:35–71.

Buckley, Eugene. 1994. Prespecification of default features: The two /i/'s of Kashaya. In *NELS 24: Proceedings of Twenty-Fourth Annual Meeting of the North East Linguistic Society*, 17–30.

Chitoran, Ioana. 2001. *The Phonology of Romanian: A Constraint-Based Approach*. Mouton de Gruyter.

Goldhahn, Dirk, Thomas Eckart, and Uwe Quasthoff. 2012. Building large monolingual dictionaries at the Leipzig Corpora Collection: From 100 to 200 languages. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation*, 759–765.

Gorman, Kyle. 2026. Greek fortition in Logical Phonology. *Canadian Journal of Linguistics* in press.

Inkelas, Sharon. 1995. The consequences of optimization for underspecification. In *Proceedings of the North East Linguistic Society 25*, 287–302.

Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69:529–574.

Inkelas, Sharon, Orhan Orgun, and Cheryl Zoll. 1997. The implications of lexical exceptions for the nature of grammar. In *Derivations and Constraints in Phonology*, ed. Iggy Roca, 393–418. Oxford University Press.

Kiparsky, Paul. 1993. Blocking in non-derived environments. In *Studies in Lexical Phonology*, ed. Sharon Hargus and Ellen Kaisse, 277–313. Academic Press.

Kodner, Jordan. 2019. Estimating child linguistic experience from historical corpora. *Glossa* 4:1–14.

Nida, Eugene A. 1949. *Morphology: The Descriptive Analysis of Words*. University of Michigan Press, 2nd edition.

Rasin, Ezer. 2025. Morpheme structure constraints solve three puzzles for theories of blocking in non-derived environments. *Linguistic Inquiry* in press.

Steriade, Donca. 2008. A pseudo-cyclic effect in Romanian morphophonology. In *Inflectional Identity*, ed. Asaf Bachrach and Andrew Nevins, 313–360. Oxford University Press.

Yang, Charles. 2005. On productivity. *Linguistic Variation Yearbook* 5:333–370.

Yang, Charles. 2016. *The Price of Linguistic Productivity: How Children Learn to Break the Rules of Language*. MIT Press.